

MULTI-RATE EXECUTION CADENCE & LOCK-FREE TRIPLE-BUFFER PROTOCOL

Decoupling Slow Deliberative Planning (20–50 Hz) from Fast Real-Time Motor Control (1000 Hz) via Hardware Sequence Locks

PANEL A: HIERARCHICAL TIME CADENCES

1. Deliberative Intent & VLA Backbone

Substrate: Linux Host / Edge NPU (Multi-modal Transformer)

Cadence: 1–5 Hz · Latency: 200–1000 ms · Non-deterministic

Emits high-level goal leases & spatial affordance bounding boxes

1–5 Hz

2. Generative Trajectory Planner (ACT / Diffusion)

Substrate: Edge GPU / NPU Kernel (Action Chunking Horizon H=16)

Cadence: 20–50 Hz · Chunk Period: $\Delta t_{\text{chunk}} = 20\text{--}50\text{ ms}$

Writes smooth trajectory chunks p_t into shared memory mailbox

20–50 Hz

3. Real-Time Safety Referee & Motor Loop

Substrate: Lock-Step MCU / FreeRTOS (Control Barrier Functions)

Cadence: 1000 Hz · Jitter: $< 5\text{ }\mu\text{s}$ · Hard Deadline: 1.0 ms

Interpolates spline waypoints; verifies dynamic stopping clearance

1000 Hz

4. Field-Oriented Current Control (FOC)

Cadence: 20–100 kHz · Hardware PWM Timers (TIMx_CCR)

Closes inner stator current loop; delivers phase voltage to windings

20–100 kHz

PANEL B: LOCK-FREE SEQUENCE LOCK PROTOCOL

SLOT 0

Writer Active Slot (seq = 2k+1)

Generative Planner writing new Action Chunk

$p_t[0..15] = \{q_k, \dot{q}_k, \tau_k\} \cdot \text{seq}++ \text{ on entry/exit}$

Writer Locked

SLOT 1

Committed Valid Mailbox (seq = 2k)

Complete, consistent chunk published for consumption

Memory Barrier: `__atomic_thread_fence(memory_order_release)` Ready to Read

SLOT 2

Reader Active Snapshot

1000 Hz Real-Time Thread evaluating CBF Barrier

`do { s1 = seq; copy(chunk); s2 = seq; } while (s1 != s2 |` Zero Wait

Sequence Lock Architectural Invariant

1. Zero Blocking: 1000 Hz safety loop NEVER blocks on slow 20 Hz writer.
2. Tearing Immunity: Atomic seq counter detects in-flight writes instantly.
3. Deterministic Fallback: If writer stalls, reader smoothly decelerates to safe stop along the valid

Reader Overhead: $< 150\text{ ns}$ · Zero Mutex Contention · Zero Heap Malloc

Triple-buffering with hardware sequence locks enables asynchronous rate decoupling across cognitive and motor silicon without lock contention or memory tearing.